ACTIVE DIRECTORY HELP DESK LAB

Detailed Project Report

A practical Windows Server and client lab focused on identity administration, access control, Group Policy, troubleshooting, and PowerShell automation.

Document field	Details
Project domain	aklab.local
Domain controller	DC01 — 192.168.50.10
Domain client	CLIENT01 — joined to aklab.local
Platform	Oracle VirtualBox; Windows Server and Windows client
Prepared	November 3, 2026
Document purpose	IT Support / Active Directory Case Study
Author	Akshveer Kaur
Version	2.0 – updated report

Document Control

Item	Description
Audience	Entry-level IT Support / Help Desk recruiters, interviewers, and technical reviewers.
Scope	A small, isolated Active Directory domain used to practise realistic support and administration tasks.
Evidence standard	Screenshots and command output should show the configuration, test, and successful result where applicable.
Security standard	No passwords, recovery codes, private tokens, or unnecessary personal information should appear in screenshots.
Status convention	Confirmed = described as working in project notes; In progress = implementation/testing is ongoing; Proposed = useful follow-up work not yet confirmed.

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1. Executive Summary

I built a small Windows Active Directory lab to practise the day-to-day work that an entry-level IT Support Technician or Help Desk Analyst may perform in a Windows domain environment. The lab uses a Windows Server domain controller named DC01 and a domain-joined Windows client named CLIENT01. The domain is aklab.local, and DC01 uses the IPv4 address 192.168.50.10.

The project focuses on identity and access management, user and group administration, organizational units (OUs), Group Policy, domain authentication, shared-folder permissions, and practical troubleshooting. I have worked through scenarios such as password resets, shared-folder access problems, domain sign-in and account-lockout concepts, employee onboarding/offboarding, delegated access, and backup-related administration.

I also use PowerShell to support repeatable administration and user communication. Alongside my Active Directory reporting toolkit, I implemented a centrally managed IT announcement workflow: a JSON announcement on DC01, a PowerShell display script, and a Group Policy Preferences scheduled task that checks for new announcements after domain user sign-in.

The purpose of this project is not to claim enterprise production experience. It is to demonstrate controlled implementation, troubleshooting, validation, and documentation. Recent work extends the lab into file recovery through Windows Server Shadow Copies, scheduled version snapshots, and a centralized user-notification workflow. These controls are documented with their limits, including that Shadow Copies are not a substitute for independent backups.

2. Objectives and Scope

2.1 Objectives

- Build a functioning Active Directory Domain Services (AD DS) environment with a domain controller and a Windows client.
- Configure and verify domain DNS and domain membership.
- Organize directory objects using OUs, users, and security groups.
- Practise user lifecycle tasks such as password resets, account status checks, onboarding, and offboarding.
- Apply and troubleshoot Group Policy settings for domain computers or users.
- Configure shared-folder access and distinguish share permissions from NTFS permissions.
- Investigate typical help desk symptoms using repeatable diagnostic steps.
- Create PowerShell scripts that produce useful administrative reports and assist with onboarding.
- Maintain clear ticket records, evidence references, and verification notes for a public-facing portfolio.
- Configure server-side Shadow Copies to expose Previous Versions for shared files.
- Schedule recurring volume snapshots and validate file-version recovery from a domain client.
- Deliver centrally authored IT announcements through PowerShell and Group Policy Preferences, with per-user acknowledgement tracking.

2.2 In scope

- A small virtualized lab isolated from production systems.
- Windows Server AD DS, DNS, domain-joined Windows client, OUs, users, groups, Group Policy, and file sharing.
- PowerShell-based read-only reporting and controlled account creation in the lab.
- Ticket documentation, screenshots, test evidence, and lessons learned.

2.3 Out of scope

- Production deployment or support for a real organization.
- High availability, multi-site replication, enterprise backup architecture, or disaster recovery validation.

- Production-grade privileged access management, identity governance, or a full Microsoft Entra ID/Microsoft 365 deployment.
- Automatic employee ID assignment through an enterprise HR workflow; this remains a separate enhancement unless implemented and tested.

Scope boundary: All the changes can only be made inside the lab.

3. Lab Environment and Requirements

Component	Configuration / role	Notes
Host computer	Windows host with 12th Gen Intel Core i5-1240P; virtualization enabled	VirtualBox is used to run the lab VMs.
Hypervisor	Oracle VirtualBox	Snapshots are recommended before major policy or directory changes.
DC01	Windows Server; domain controller and DNS server	IPv4: 192.168.50.10.
Domain	aklab.local	Internal lab domain name.
CLIENT01	Windows client joined to aklab.local	Used to test domain sign-in, Group Policy, and access to shares.
PowerShell working folder	C:\Powershell IT Scripts	Current folder used for the administration scripts.
Shared resource	\\DC01\CompanyShare	Shared-folder permissions have been configured and tested in the lab.
File-version service	Windows Server Shadow Copies on DC01 system volume	Configured with a 5 GB shadow-storage limit in the lab; shared-file Previous Versions tested.
Announcement repository	C:\CompanyShare\IT-Announcements	Current-Announcement.json and Show-ITAnnouncement.ps1; read access for intended users.
Notification delivery	GPO: AKLab-IT-Announcements-Policy	Group Policy Preferences scheduled task; 30-second logon delay.

3.1 Prerequisites

- Virtualization enabled in firmware and enough host storage and memory for both virtual machines.
- Windows Server installed and promoted to a domain controller with AD DS and DNS.
- CLIENT01 configured to use the domain controller for DNS and successfully joined to aklab.local.
- A domain account with appropriate rights for directory changes; local elevation alone does not automatically grant domain permissions.
- The ActiveDirectory PowerShell module available on the system where scripts are run.

3.2 Useful environment checks

```
ipconfig /all
nslookup aklab.local
nltest /dsgetdc:aklab.local
Get-Module -ListAvailable ActiveDirectory
Get-ADDomain
```

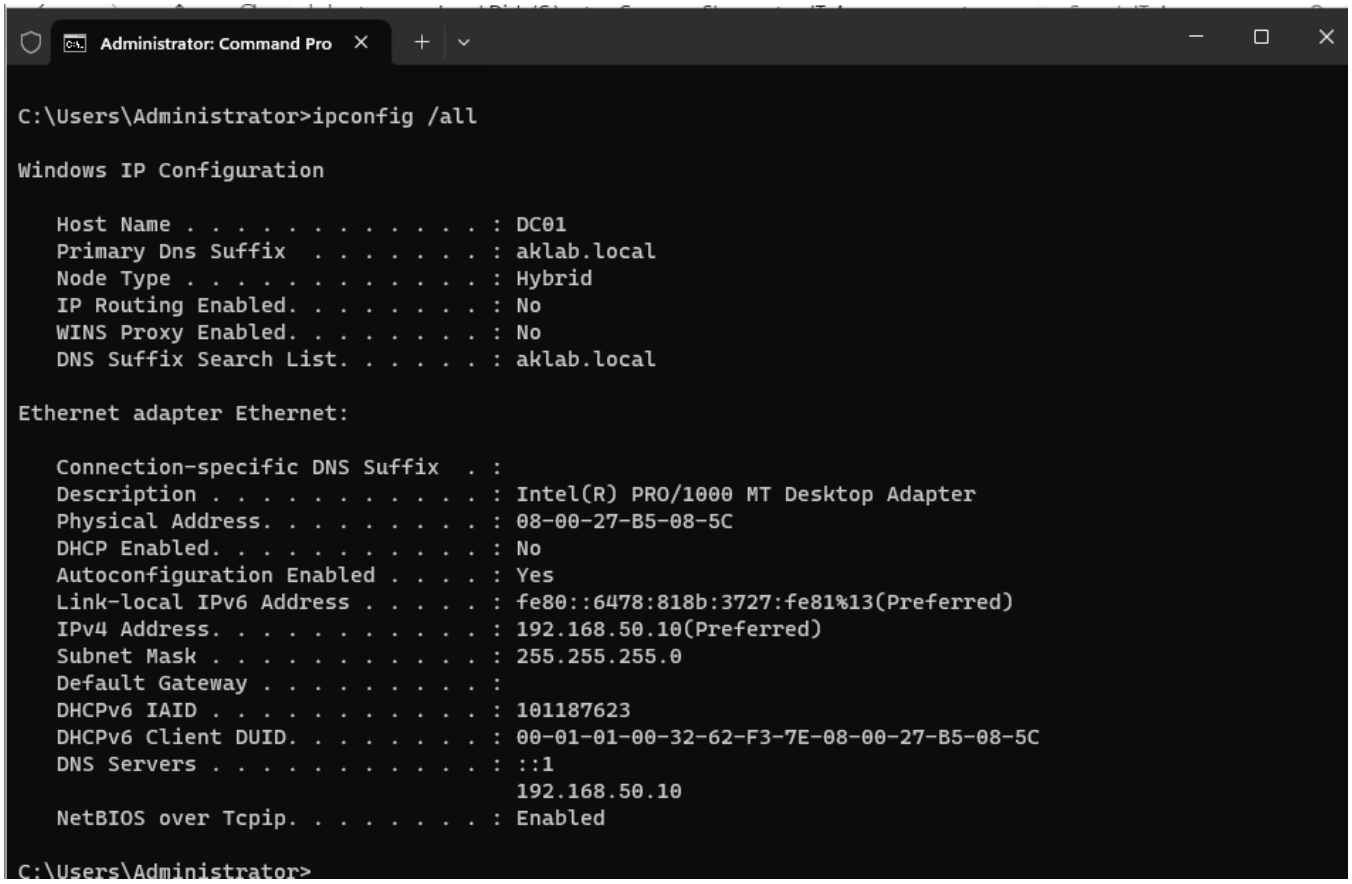


Figure 1: ipconfig /all

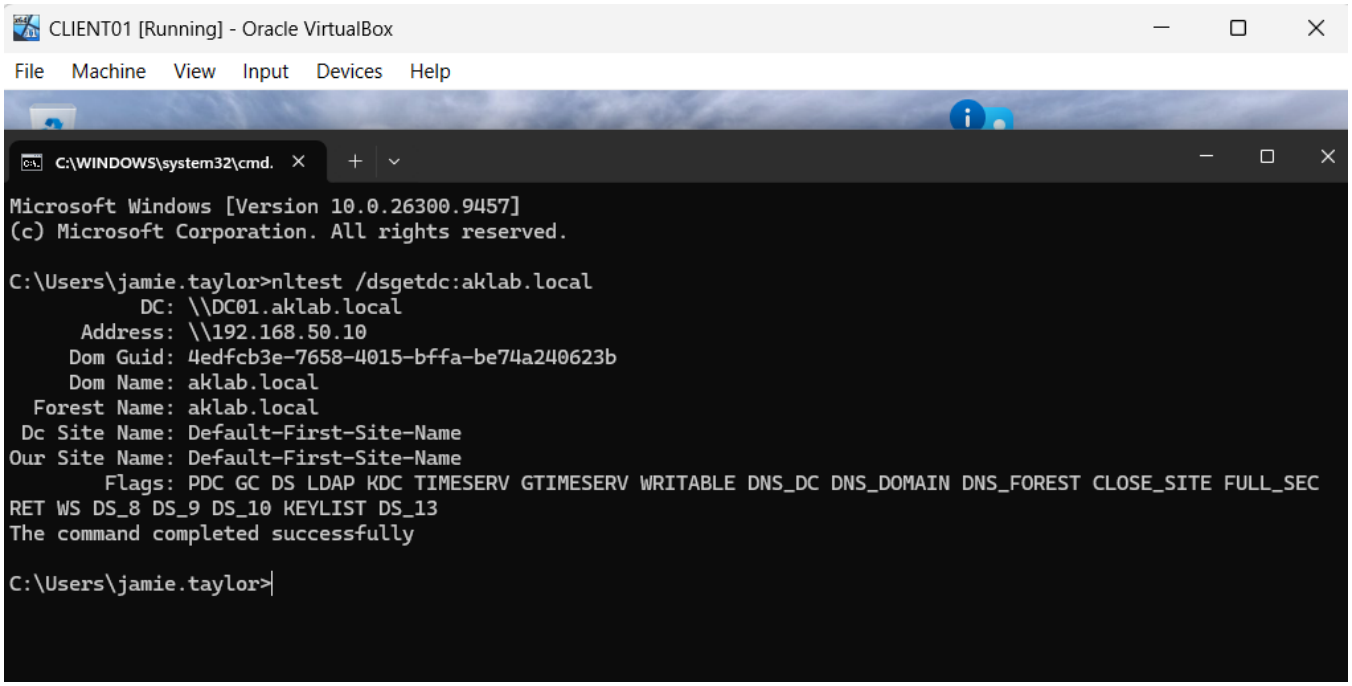


Figure 2: The connection is built between DC01 and CLIENT01

4. Network and Domain Architecture

The lab uses a simple two-machine architecture. DC01 provides Active Directory Domain Services and DNS for the aklab.local domain. CLIENT01 is the workstation used to validate domain authentication, Group Policy processing, and access to shared resources.

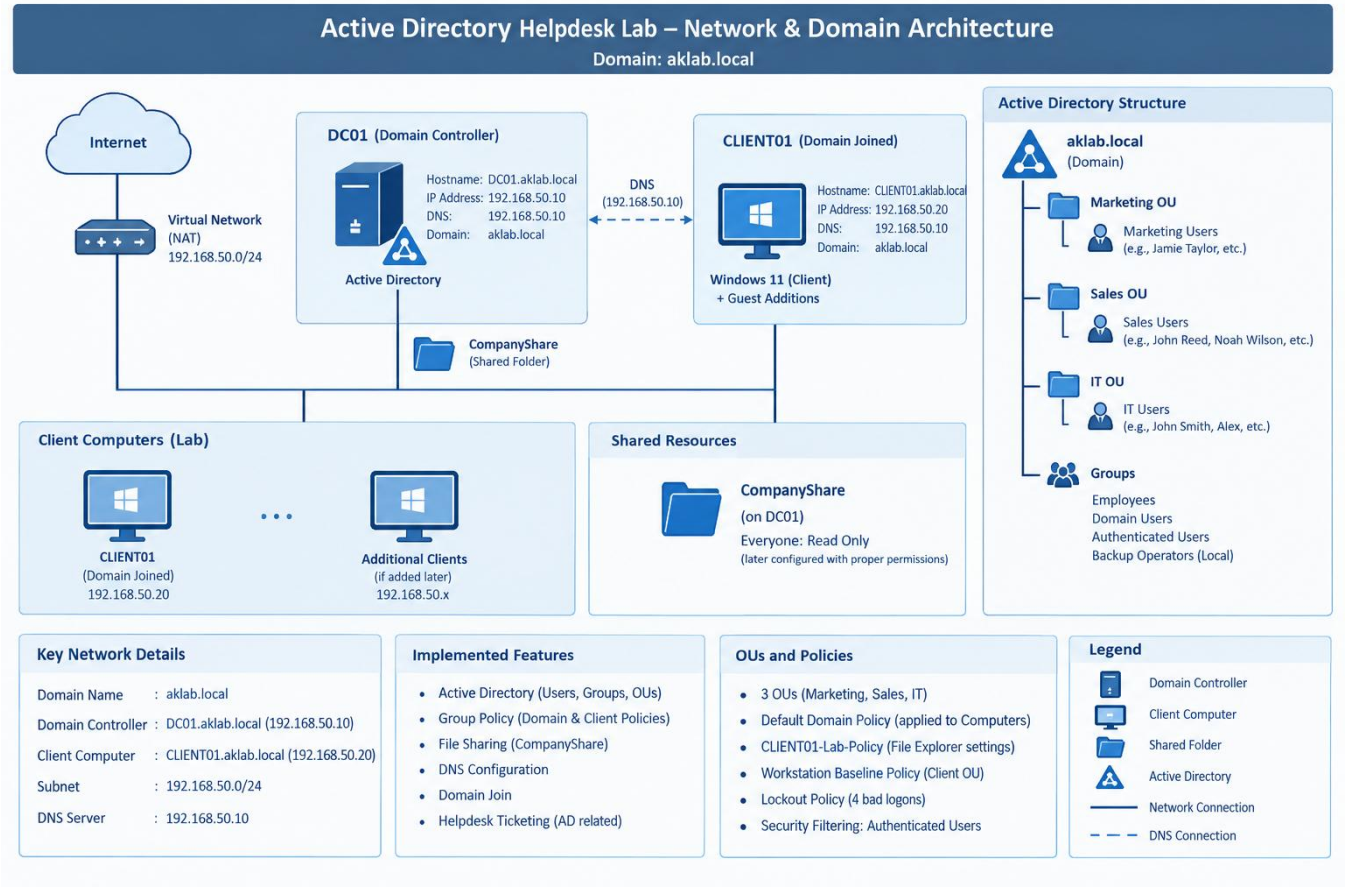


Figure 3: Lab Architecture Diagram

Element	Purpose	Validation
DC01	Central identity directory, domain authentication, DNS, and lab shared folder host.	Confirm domain services and DNS are functioning; run AD cmdlets successfully.
192.168.50.10	Static IPv4 address recorded for DC01.	Use ipconfig /all on DC01 and verify the client can resolve/reach it.
DNS	Allows the client to locate domain services using AD DNS records.	Confirm CLIENT01 points to the lab DNS server and resolves aklab.local.
aklab.local	Active Directory domain namespace.	Use whoami /upn or domain-related commands on CLIENT01.
CLIENT01	Domain-joined endpoint for client-side testing.	Run gpresult and test domain sign-in and share access.
\\DC01\CompanyShare	Network share for validating file access.	Test using the intended user account and document allowed/denied access.

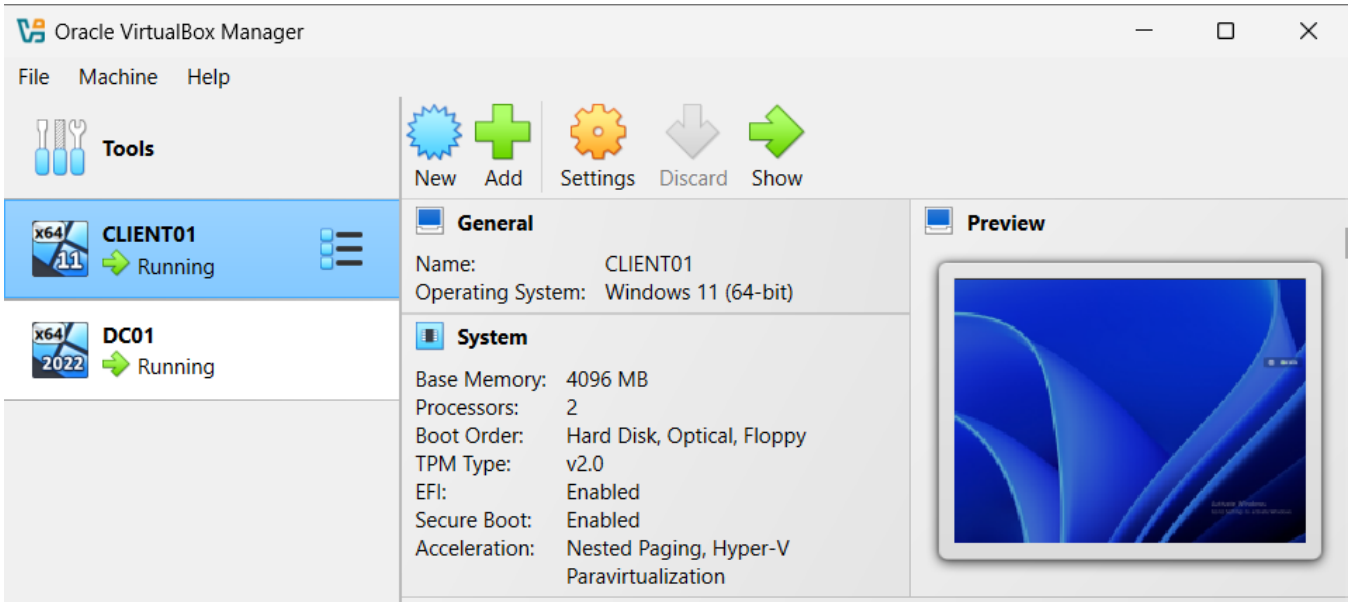


Figure 4: DC01 and CLIENT01

4.1 DNS and domain-join troubleshooting

During setup, the client's DNS configuration required attention before domain joining and domain resource access worked reliably. This is a useful example of why domain-join issues should not be treated as password problems until name resolution and domain controller discovery have been checked.

1. Check the client's IP address, subnet, default gateway, and DNS server with ipconfig /all.
2. Verify that the client uses the lab DNS server rather than an unrelated external DNS server for the AD domain.
3. Use nslookup aklab.local and nltest /dsgetdc:aklab.local to test name resolution and domain controller discovery.
4. After correcting DNS, retry the domain operation and validate sign-in on CLIENT01.

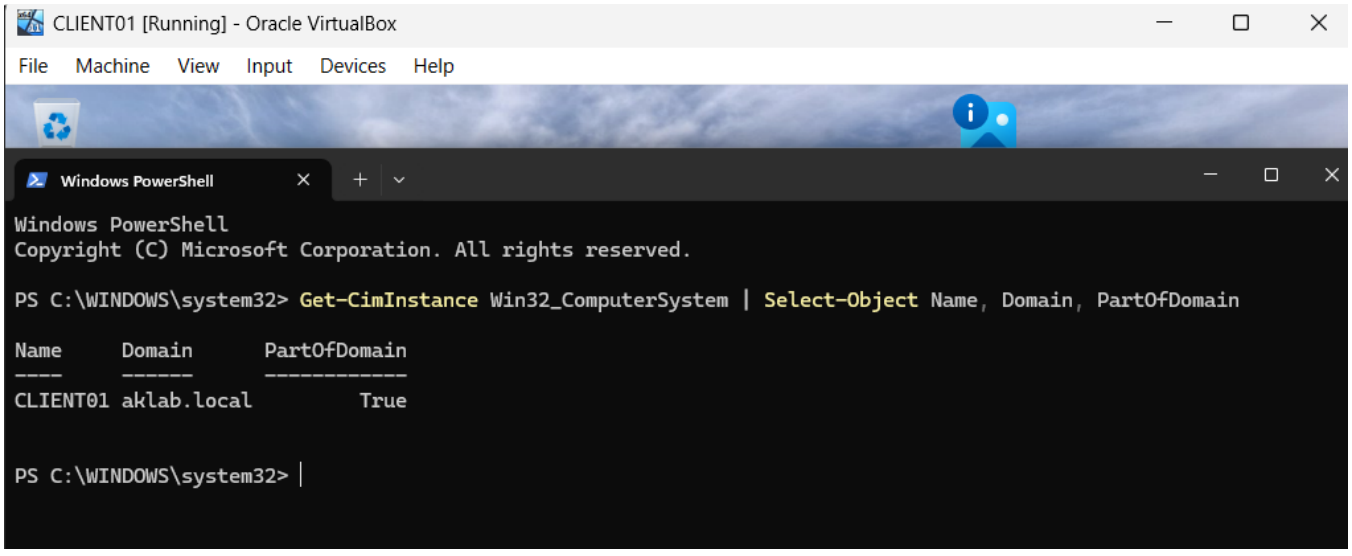


Figure 5: CLIENT01 is joined to the domain aklab.local

5. Active Directory Design

5.1 Organizational Units

I have used OUs to organize directory objects and provide logical locations where Group Policy can be linked. The lab includes OUs for areas such as HelpDesk, Employees, and Workstations, with additional departmental organization such as Marketing considered as the lab develops. The exact OU list should be confirmed against the current directory before publishing this report.

Directory object	Purpose	Design note
HelpDesk OU	Organize help desk-related user or computer objects.	Use a clear naming convention and document the objects stored here.
Employees OU	Organize employee user accounts.	OU placement does not itself grant file access.
Workstations OU	Organize computer accounts such as CLIENT01.	Can be used as a target for computer configuration GPOs.
Departmental OUs	Optional organization for teams such as Marketing or Sales.	Use only if it improves management; avoid creating unnecessary OU complexity.

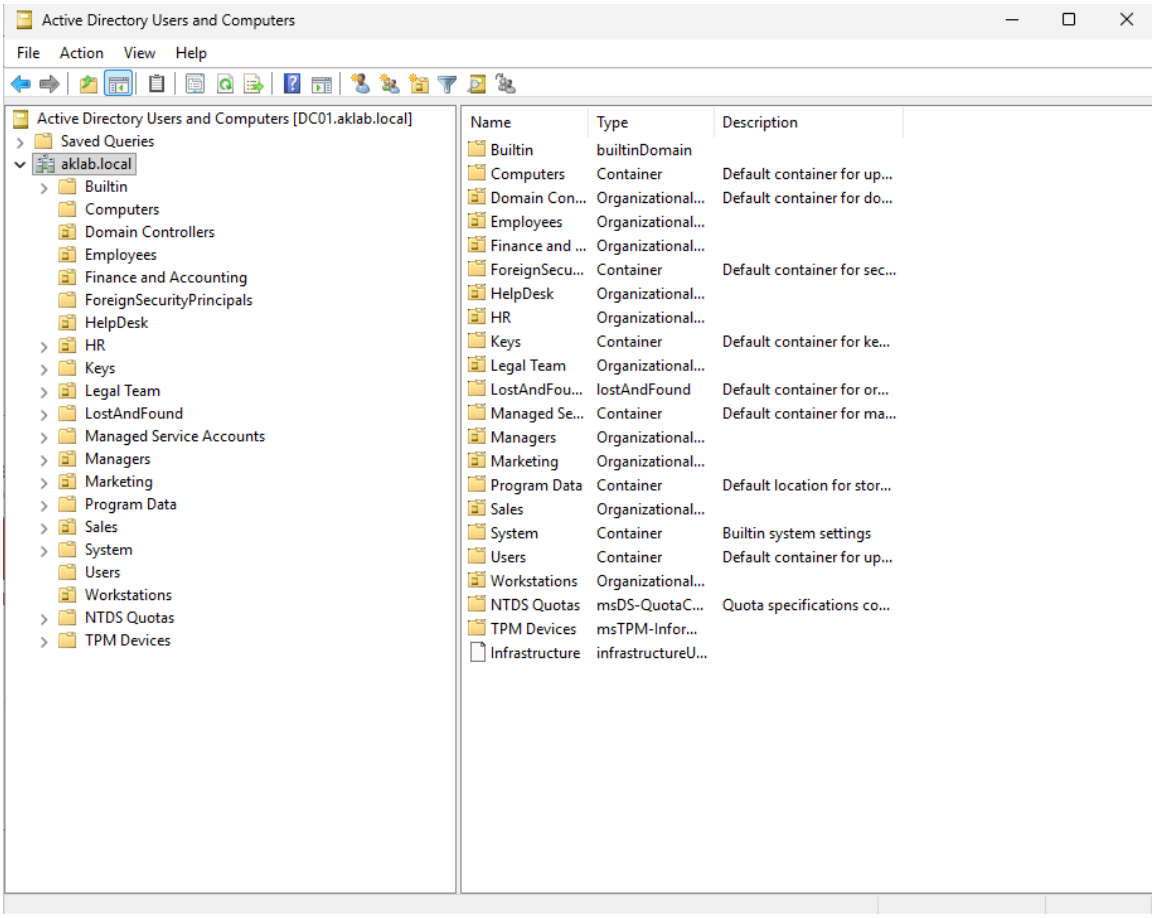


Figure 6: Organizational Units inside aklab.local

5.2 Users and security groups

The lab contains test user accounts used for support scenarios, including Alex Morgan, Jamie Taylor, John Reed, John Smith, and Noah Wilson. These accounts represent different administrative situations, such as shared-folder access, manager relationships, backup-related responsibilities, and employee offboarding.

A user account identifies an individual. A security group can be used to assign permissions to a set of users. An OU organizes objects and can be a scope for Group Policy. These concepts are related but not interchangeable: creating users inside a Marketing OU does not automatically create a Marketing security group or grant Marketing-specific share permissions.

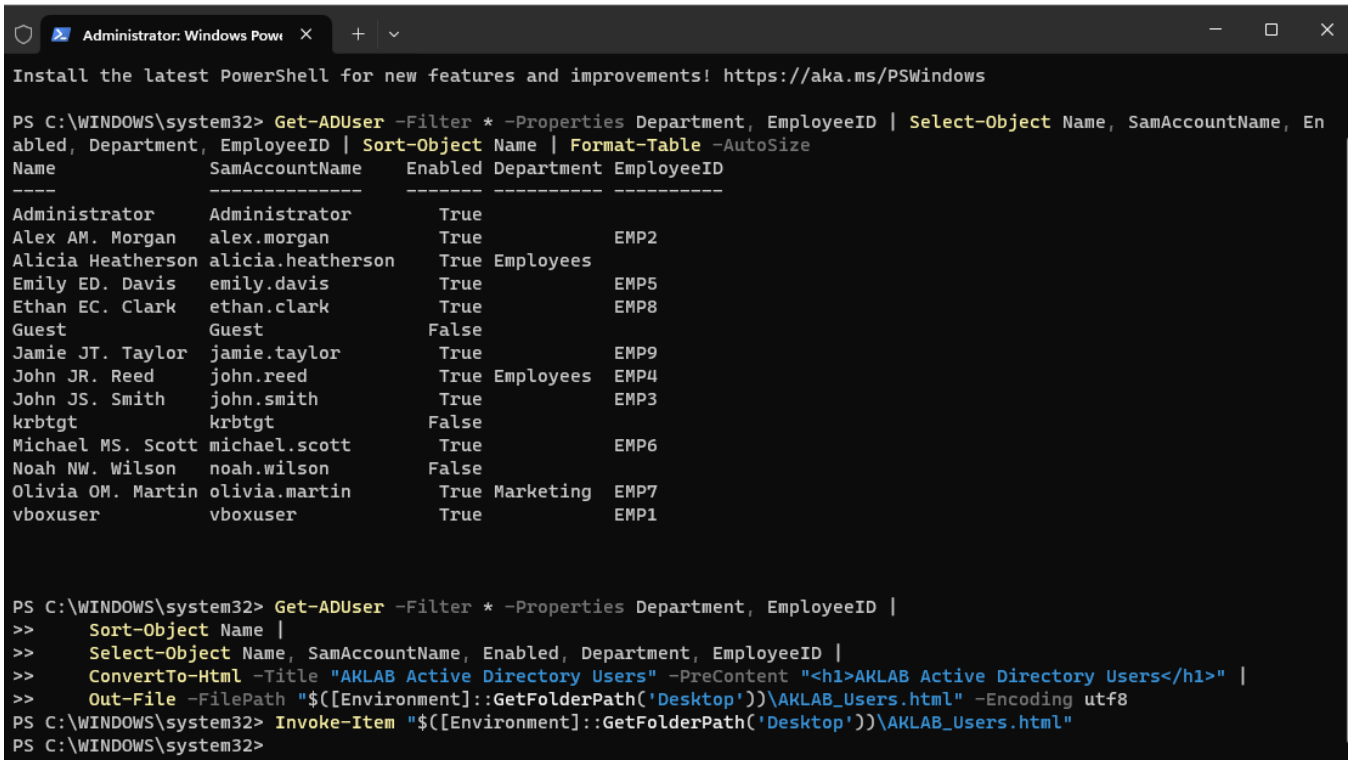


Figure 7: List of the employees and generating an employees report

Refer to this report: AKLAB_Users.html

5.3 Naming and administration principles

- Use consistent account and group naming conventions.
- Grant only the access required for the job role and validate it using the affected test account.
- Prefer group-based access assignments over repeatedly assigning permissions to individual users.
- Check the exact group name in AD before running scripts that query a named group.
- Document manager relationships and employee identifiers only after verifying the relevant AD attributes and organizational requirements.

6. Implementation and Configuration

6.1 Domain and DNS configuration

I configured DC01 as the domain controller for aklab.local and set its recorded IPv4 address to 192.168.50.10. DNS was configured to support domain discovery, and CLIENT01 was subsequently joined to the domain. I validated domain sign-in after troubleshooting the client's DNS configuration.

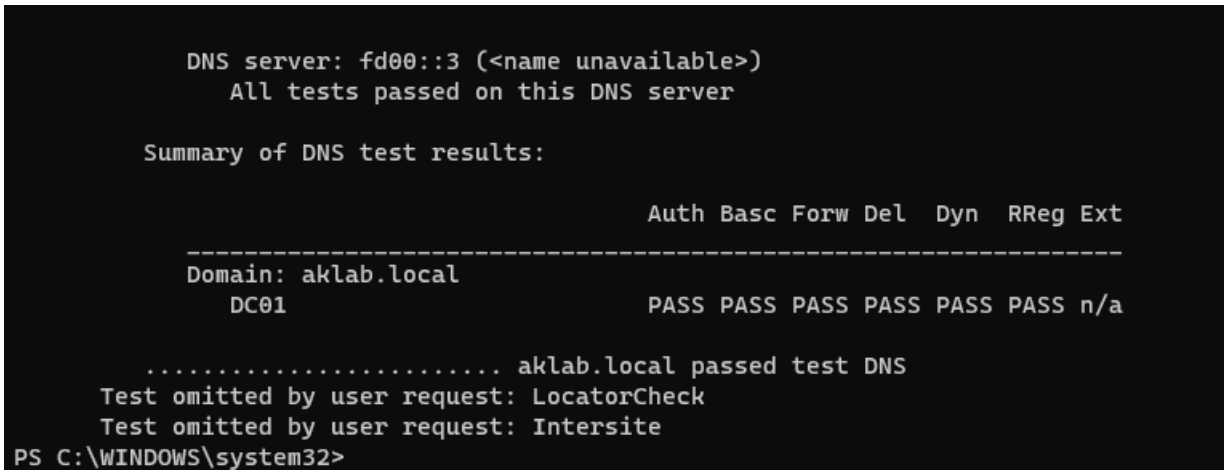


Figure 8: DNS Heath Check Result – DNS passed all six displayed DNS checks

6.2 Group Policy

I created and tested Group Policy settings for the lab, including a policy named CLIENT01-Lab-Policy and a workstation baseline policy. The CLIENT01-Lab-Policy link was associated with the lab domain/target structure and had the recorded GPO ID {8A1830F1-8574-4B8C-B9D0-FCA47AD9D617}. During troubleshooting, gpresult initially showed policy processing as inaccessible, empty, or disabled; this was investigated, and the policy was later reported as applying.

I also reviewed the Default Domain Policy while configuring account lockout settings. In a domain, account policy settings such as password and lockout policies are typically managed through domain-linked policy settings; the effective policy should always be verified rather than assuming that a setting has applied because it was edited.

```
gpupdate /force
gpresult /r
gpresult /h "$env:USERPROFILE\Desktop\GPResult.html"
```

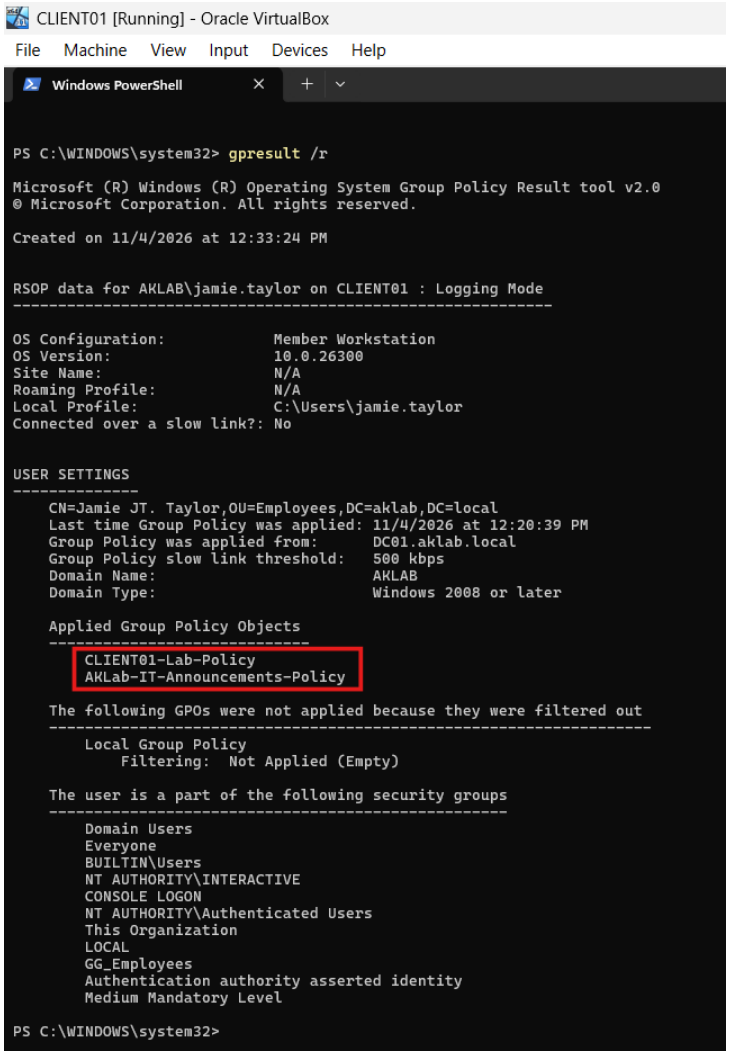


Figure 9: Applied group policies to a user Jamie Taylor's account

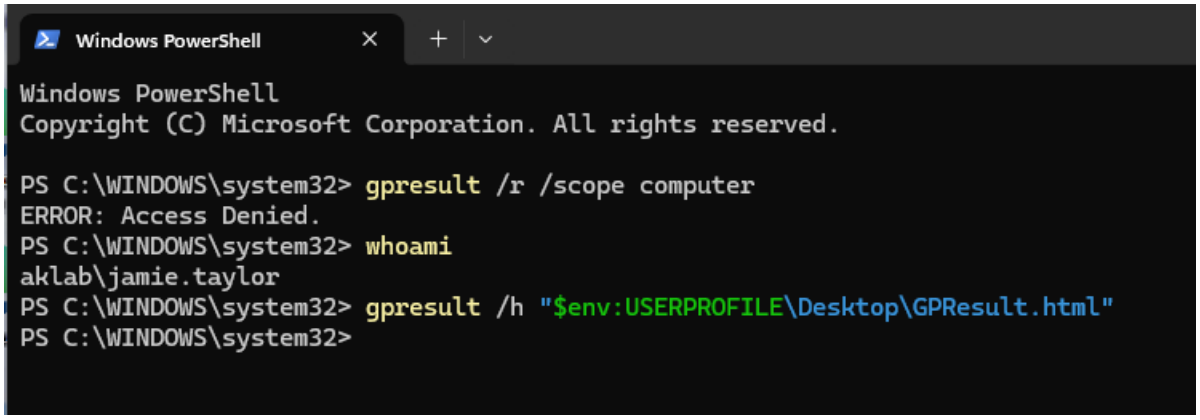


Figure 10: The user is denied access of viewing applied computer policies for security reasons

GpResult Report is here.

[Reports\GPResult.html]

6.3 Shared folders and permissions

I configured the CompanyShare network share on DC01 and worked through access problems involving paths such as \\192.168.50.10\CompanyShare and \\DC01\CompanyShare. The initial share permissions included Everyone with read access, while folder-level access was also affected by NTFS permissions and inherited groups. The access issue was later reported as resolved, and testing indicated that users could access the folders intended for them.

The key lesson is that effective network-share access depends on both share permissions and NTFS permissions. When troubleshooting, I need to inspect both layers, test using the affected user account, and verify both expected access and expected denial. A user successfully opening a folder does not, by itself, prove that other users are correctly blocked.

1. Confirm the share exists and the UNC path is correct.
2. Review share permissions and NTFS Security-tab permissions separately.
3. Check inherited permissions and group memberships.
4. Test access while signed in as the intended domain user.
5. Record both successful access and a negative test where access should be denied.

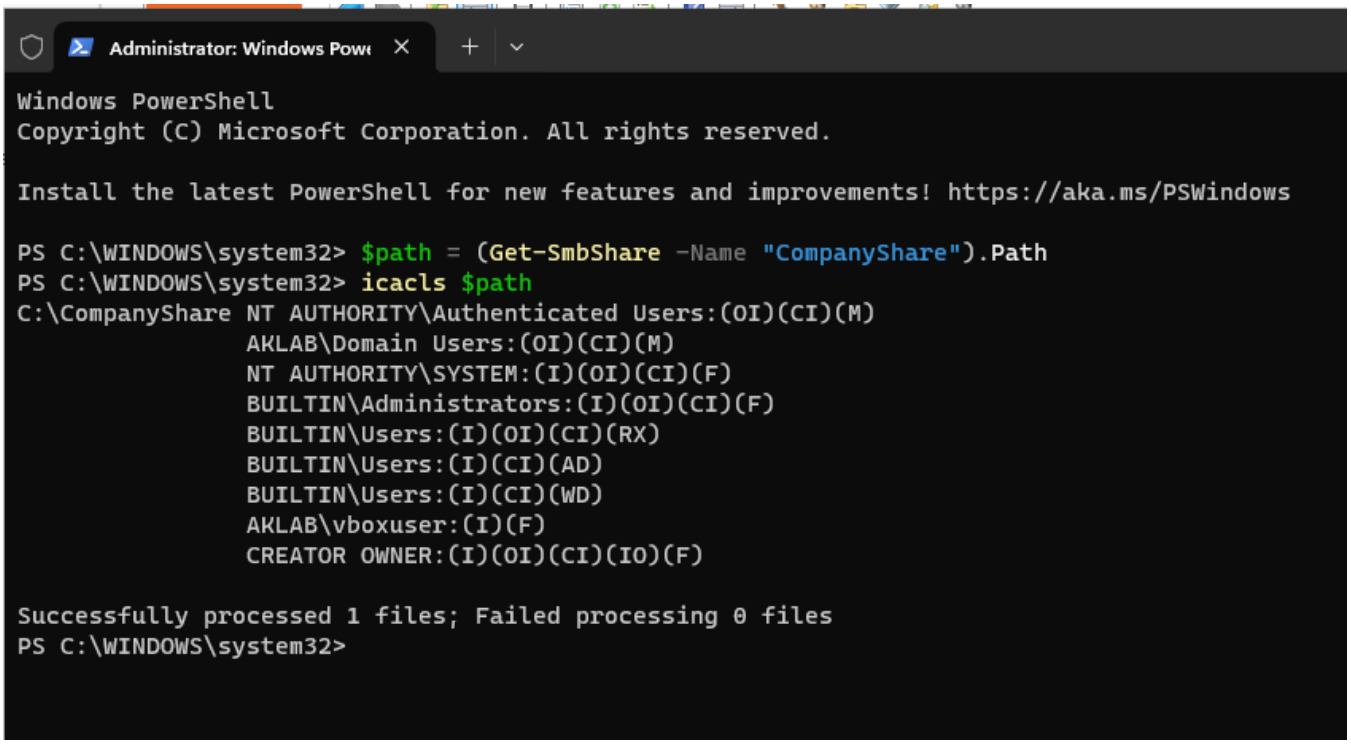


Figure 11: NTFS Permissions

6.4 Account lifecycle and access administration

The ticket scenarios cover common identity tasks: password resets, investigating domain login failure, reviewing account lockout behavior, onboarding a new employee, and disabling an account during offboarding. For offboarding, the safe initial action in this lab is to disable the account and review access and ownership; deletion should not be automatic because business records and ownership may need to be retained.

The project also considers employee IDs in a format such as EMP001. A robust implementation must check existing IDs before assigning the next available value and must prevent duplicates. The automatic assignment of employee IDs is a separate automation feature and should only be described as complete after it has been implemented and tested against existing accounts.

6.5 Backup-related access

A lab scenario explored whether a user assigned backup-related responsibilities should have access to the Administrator account or password. The safer principle is that the user should not be given another person's administrator credentials. Access should instead be granted through an appropriate, limited role or a controlled backup process. Any backup operator group membership should be validated for the actual machine and context, because a local built-in group and an AD domain group are not the same thing.

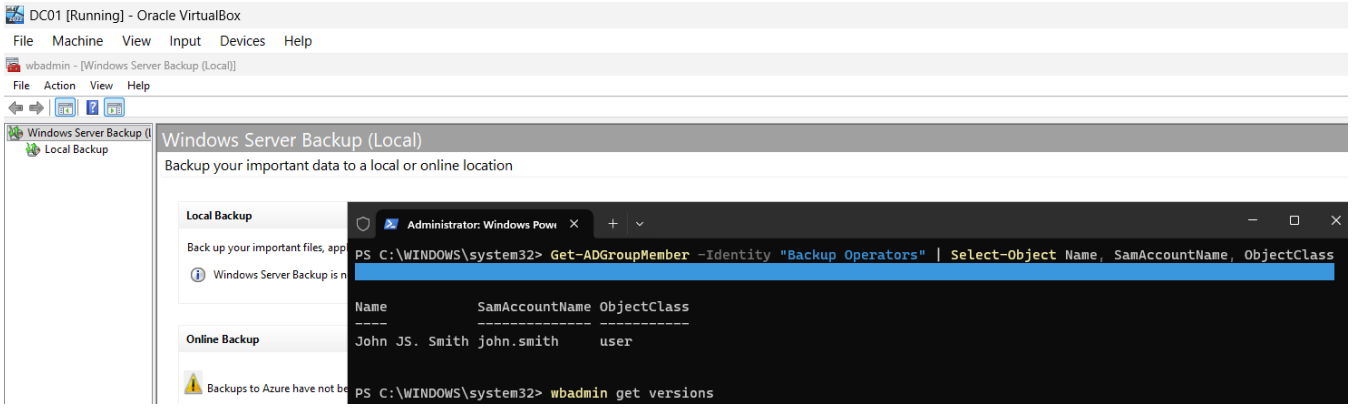


Figure 12: Backup related access and information

6.6 Shadow Copies and File Version History

Business requirement: Employees working in the CompanyShare folders needed a way to retrieve an earlier version of a document after an accidental overwrite or unwanted edit, without requiring a full backup restore for each incident.

Implementation: On DC01, I enabled Windows Server Shadow Copies for the volume hosting CompanyShare (C: in this lab). I set a 5 GB shadow-copy storage limit and created an initial snapshot. I used a controlled test document, Employee-Policy-Test.txt, in the Employees shared folder to exercise the Previous Versions feature.

Validation scenario: I first saved a Version 1 of the test document, captured a shadow copy, and then signed in to CLIENT01 as Jamie Taylor. I changed the document to Version 2 and inspected the file/folder Properties > Previous Versions interface to identify the earlier snapshot. This tested the relationship between the server-side snapshot and the client-facing recovery interface.

Recovery approach: For a real support request, I would inspect the available previous versions, open or copy the desired version to a safe location, compare contents, and obtain approval before overwriting the current file. A restore operation can replace later changes and should be treated as a controlled action.

Technical boundary: Shadow Copies operate at the volume level; they are not individual per-folder backups. Available versions depend on the snapshot schedule, retention/storage limits, and whether a snapshot existed before the change. They do not protect against loss of the underlying volume.

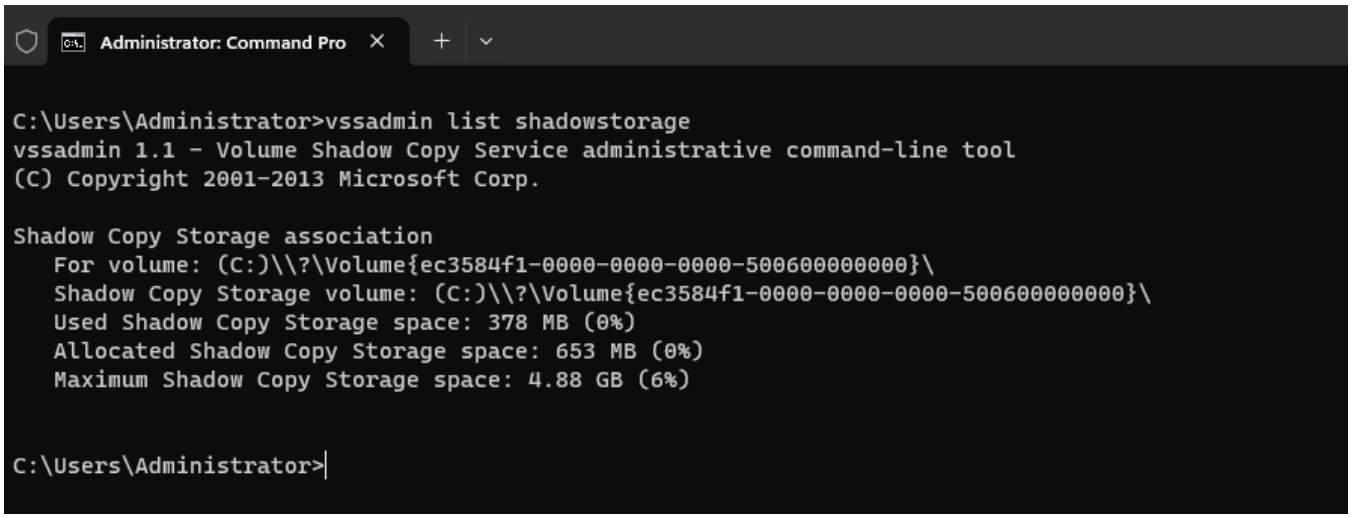


Figure 13: Shadow Copy Information

6.7 Automated Version History — HD-023

Business requirement: A manually created shadow copy provides only a single recovery point. I therefore extended the file-version history configuration to support automatic, recurring snapshots without requiring an administrator to initiate every capture.

Implementation design: I configured or worked through the Shadow Copies scheduling interface on DC01, using twice-daily snapshots at 9:00 AM and 3:00 PM as the planned lab schedule. The schedule belongs to the volume's shadow-copy configuration; users do not need a client-side script to create snapshots.

Operational checks: The relevant tests are to confirm that the scheduled snapshot task exists, inspect its next and last run times, wait for or trigger a controlled scheduled capture, and verify that a newly created recovery point appears in Previous Versions. A task merely being enabled is not proof that scheduled snapshots are completing.

Retention considerations: The 5 GB allocation constrains how many recovery points may remain available. Older shadow copies can be removed when storage pressure requires it. Snapshot frequency and storage allocation should be reviewed against the actual file-change rate and recovery expectations.

Validation status: Manual snapshot creation and file-version testing were discussed and exercised in the lab; final unattended execution at both scheduled times should be evidenced with task history and distinct snapshot timestamps before describing the schedule as independently verified.

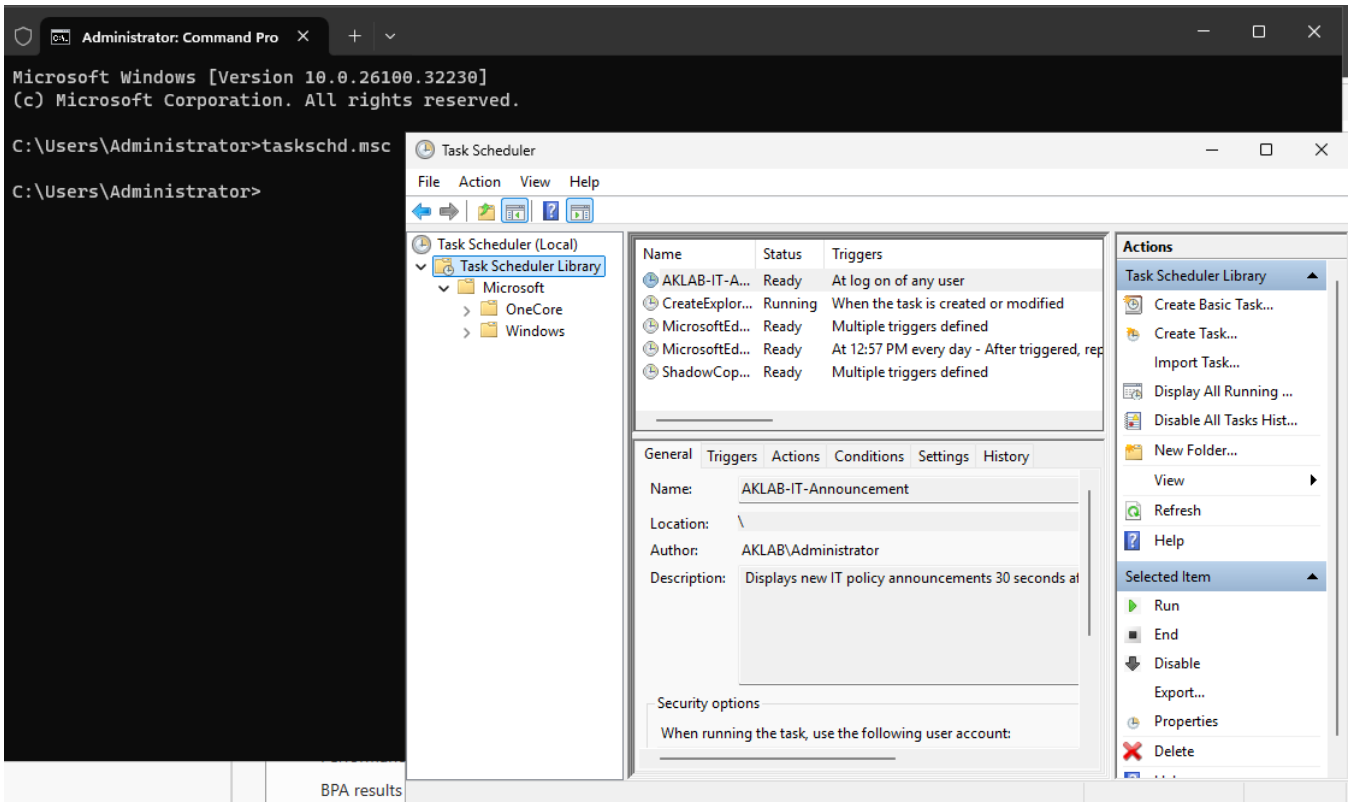


Figure 14: Shadow Copy Task Scheduler

6.8 Centralized IT Announcement System — HD-024

Business requirement: When IT introduces a new policy or service, employees need a consistent way to receive an announcement without an administrator editing each workstation or user profile individually. I used the new file-version history policy as the first practical notification scenario.

Central content: On DC01 I created C:\CompanyShare\IT-Announcements and stored Current-Announcement.json as the single current notice. The JSON includes AnnouncementID, Title, Message, PublishedBy, and Version. Administrators can publish a new notice by updating the JSON and assigning a new unique AnnouncementID.

PowerShell delivery: Show-ITAnnouncement.ps1 reads the JSON through \\DC01\CompanyShare\IT-Announcements\Current-Announcement.json, checks whether the current announcement ID is present in the signed-in user's local state file, and displays a Windows Forms message box when it is new. After the user clicks OK, the script records the ID under %LOCALAPPDATA%\AKLabIT\SeenAnnouncements.txt. Errors can be written to AnnouncementErrors.log in the same per-user folder.

Deployment: I created AKLab-IT-Announcements-Policy and configured a Group Policy Preferences scheduled task named AKLAB-IT-Announcement. Its logon trigger is enabled with a PT30S (30-second) delay. The task launches powershell.exe in the interactive user context and points to the centrally maintained .ps1 file. The pop-up appears on the user desktop after sign-in, rather than as a Windows toast or pre-login message.

Troubleshooting and resolution: During testing, the task was initially available for Jamie Taylor but not consistently for Olivia Martin. I compared gpresult, task enumeration, trigger configuration, principal details, and Task Scheduler results. Group Policy Preferences logged Event ID 4098 with 0x80070005 (Access is denied), highlighting the distinction between a GPO applying and its preference item successfully creating a task. I reviewed the Common-tab security-context option and the task registration/security configuration. The final working arrangement was reported as functioning across the tested accounts.

Validation: I manually launched the script, confirmed the scheduled task could launch it, inspected the logon trigger (Enabled=True, Delay=PT30S), and tested delivery with multiple user accounts. I verified the design for one-time display by assigning a new AnnouncementID for each publication. The scheduled task and announcement process were subsequently reported as working correctly.

Security model: Employees require read access to the announcement files, while only administrators should be able to modify the JSON or script. Authenticated Users read permission is suitable only for non-confidential company-wide notices. Running an untrusted or writable PowerShell script from a network share would be unsafe; the script source must remain administrator-controlled.

Operational limitation: This design checks at sign-in, not continuously. A user already signed in may not see a newly published notice until the next sign-in; per-user state is local to each Windows profile and does not provide centralized proof of reading. Dismissing the pop-up is not equivalent to acknowledging or understanding a policy.

For more information, refer to HD-024.

7. Help Desk Ticket Workflow and Case Studies

I used tickets to turn each lab task into a small, repeatable support case. Each ticket should record the reported issue or request, the affected user or device, the investigation, the confirmed root cause, the resolution, validation, and evidence. If the root cause isn't yet confirmed, label it as a hypothesis rather than stating it as fact.

7.1 Standard ticket structure

Field	What I record
Ticket ID and title	A unique ID such as HD-001 and a short, searchable title.
Request / impact	What the user needs, what is failing, and the practical impact.
Environment	User, computer, domain, share, GPO, or script involved.
Priority and status	A reasonable lab priority and whether the ticket is Open, In Progress, Resolved, or Closed.
Investigation	Commands, settings inspected, symptoms, and relevant output.
Root cause	The confirmed technical reason, or explicitly marked as unconfirmed.
Resolution	The change made and why it addresses the issue.
Validation	A test performed as the affected user/device to prove the result.
Evidence	Screenshot/output file names and a short caption for each.
Prevention / follow-up	Documentation, monitoring, permissions review, or automation that may reduce recurrence.

7.2 Known ticket scenarios

Ticket / scenario	Purpose and current documentation guidance
Password Reset Request	Document identity verification, reset process, temporary password handling, forced change at next logon if appropriate, and successful sign-in. Never include the password in evidence.
Jamie Taylor Cannot Access Employees Shared Folder	Record the reported access failure, review group membership and share/NTFS permissions, apply the minimum required access, and validate as Jamie. Project notes indicate access was later working.
Domain login failure	Check domain username format, network connectivity, DNS, domain controller discovery, account status, and time-related issues before assuming the password is wrong.
Account lockout after repeated failed attempts	Verify the lockout policy and effective settings, reproduce safely with a test account, unlock the account if appropriate, and confirm successful sign-in.
Offboarding — Noah Wilson (Sales)	Disable the account, review group memberships and access, document ownership/retention needs, and confirm that sign-in is blocked. Do not delete the account without a retention decision.
Management folder access for John Reed	Confirm the intended business requirement, use a suitable security group, apply least-privilege access, and test both allowed and denied access.
Backup-related access for John Smith	Avoid sharing Administrator credentials. Validate the correct group and privilege

	scope, then test the intended backup task with an appropriately authorized account.
Employee ID assignment	Use an agreed format such as EMP001; check existing IDs for uniqueness and test duplicate prevention. Treat automatic assignment as a separate feature until validated.
Marketing software request (7-Zip example)	For the current two-VM lab, test installation on CLIENT01 and verify it is available to domain users on that computer. Document that this is per-computer deployment, not deployment to multiple separate endpoints.
Develop PowerShell Scripts for Active Directory Administration	Track the creation, testing, error handling, and evidence for the five administrative scripts described in Section 8.
Configure File Version History (Shadow Copies)	Enabled volume-level Shadow Copies on DC01 for CompanyShare; created a controlled earlier-version test and reviewed Previous Versions from CLIENT01. Document snapshot/restore proof and the distinction from backup.
Automated Version History	Configured the planned recurring snapshot schedule (9 AM and 3 PM), reviewed retention/storage implications, and identified task-history and timestamp evidence required to verify unattended runs.
Centralized IT Announcement System	Created a JSON-driven PowerShell notification, deployed an interactive logon scheduled task via GPO with a 30-second delay, troubleshot 0x80070005 / Event 4098, and tested delivery for multiple domain users.

7.3 Evidence naming convention

I will use consistent file names so evidence can be located quickly from the main report or ticket register.

```
HD-001 PasswordResetValidation.docx
HD-002 SharePermissions.docx
HD-003 GPResultValidation.docx
HD-004 GroupMembershipReport.docx
HD-005 OUInventory.docx
```

```
Standard convention used:

        HD-xxx FileName.docx

        HD-XXX FileName.pdf
```

To get more information, refer to the ticket register.

8. PowerShell Administration Toolkit

I built five separate PowerShell scripts to practise repeatable Active Directory administration. The scripts should be run in the lab with an account that has the required permissions. Running PowerShell as Administrator elevates the local process but does not automatically grant all domain permissions. The master scripts can be accessed from here.

Script	Purpose	Expected output / validation
01-Get-ADUserReport.ps1	Export an inventory of AD users with fields such as name, username, department, email, and enabled status.	AD-User-Report.csv; compare sample rows with ADUC.
02-Find-DisabledAccounts.ps1	Find disabled user accounts and export relevant attributes, including department and last-change information.	Disabled-Accounts.csv; verify against a known disabled test account or directory status.
03-Get-GroupMembershipReport.ps1	Export members of a specified AD group, including nested membership when using -Recursive.	Group-Membership-Report.csv; use the exact existing group name and compare with ADUC.
04-Export-OUInventory.ps1	Export OUs and useful attributes such as distinguished and canonical names.	OU-Inventory.csv; compare with the actual OU structure.
05-New-EmployeeAccount.ps1	Interactively collect employee details, validate required fields and username uniqueness, verify the target OU, request an initial password securely, and create the account after confirmation.	Create a clearly identified test account in a test OU; verify its attributes and enabled state; remove/disable test objects according to the lab cleanup plan.

8.1 Script execution and testing process

1. Take a VirtualBox snapshot before making significant changes.
2. Open PowerShell on DC01 and confirm that the ActiveDirectory module is available.
3. Change to the script folder and run one script at a time.
4. Review the console output and inspect the generated CSV on the profile Desktop.
5. Compare the report with Active Directory Users and Computers or a direct AD query.
6. Test normal conditions and at least one useful edge case, such as an empty group, an unknown group name, or an already-existing username.
7. Capture evidence and update the ticket with the actual result, including any errors encountered and fixed.

```
Import-Module ActiveDirectory -ErrorAction Stop
Get-ADDomain
Set-Location "C:\Powershell IT Scripts"
.\01-Get-ADUserReport.ps1
```

8.2 Lessons from script troubleshooting

During testing, PowerShell reported that -Properties was not recognized when a parameter was separated from the Get-ADUser command by an incorrect line break. The fix was to keep the parameter on the same command line or use a valid PowerShell line continuation. The group membership report also initially failed because the group name Employees could not be found under DC=aklab,DC=local. This reinforced the need to query the directory for the real group name instead of assuming a group exists. The report count should be calculated explicitly, for example with $MemberCount = @($Members).Count, so the output is consistent when there are zero, one, or multiple members.

```
Get-ADGroup -Filter * | Select-Object Name, SamAccountName
$MemberCount = @($Members).Count
```

Security and quality: The passwords must not be stored or captured in scripts or screenshots.

8.3 Announcement automation and maintenance

Unlike the AD reporting scripts, the announcement script runs in a standard domain user session and does not require domain administrator credentials. The scheduled task launches the script, which reads a centrally maintained JSON payload and writes only per-user acknowledgement state to the local profile.

```
Get-ScheduledTask -TaskName "AKLAB-IT-Announcement"

Get-ScheduledTaskInfo -TaskName "AKLAB-IT-Announcement"

Get-Content "$env:LOCALAPPDATA\AKLabIT\SeenAnnouncements.txt"
```

Publishing procedure: edit Current-Announcement.json on DC01, change AnnouncementID to a never-before-used value, update Title and Message, validate JSON syntax, and save. Do not edit the GPO or .ps1 file for routine announcement changes. Archive previous notices separately for change history.

9. Testing and Validation Plan

For every ticket or script, I will record a test that proves the expected outcome. Validation should be specific: for example, testing share access as Jamie Taylor is stronger evidence than showing only a permissions dialog.

Area	Test	Evidence of success
Domain/DNS	Confirm client DNS points to DC01 and the client can discover the domain controller.	ipconfig /all, nslookup, nltest output.
Domain sign-in	Sign in to CLIENT01 with a lab domain account.	whoami /upn or a screenshot of the signed-in domain user, with no password shown.
Group Policy	Refresh policy and review the effective result on CLIENT01.	gpresult /r or HTML report; relevant policy

		settings.
Share access	Test access as the intended user and test denial for a user who should not have access.	Screenshots/output of both positive and negative tests.
Account lockout	Check the effective lockout policy and reproduce carefully with a test account.	Policy settings, account status, and successful post-unlock sign-in.
Offboarding	Disable the test user and verify that authentication is blocked.	AD account status and failed sign-in test without exposing credentials.
PowerShell reports	Compare CSV contents and row counts with AD queries or ADUC.	CSV file plus console summary.
Account creation	Create a uniquely named test user in the intended OU and verify attributes.	ADUC or Get-ADUser output; no password displayed.
Account creation	Create a uniquely named test user in the intended OU and verify attributes.	ADUC or Get-ADUser output; no password displayed.
Shadow Copies	Create a baseline snapshot, edit a test file as Jamie, inspect Previous Versions, and copy/restore a known earlier version safely.	Snapshot timestamp, file before/after comparison, Previous Versions window, verified recovered content.
Automated snapshots	Inspect schedule and compare snapshots produced by separate unattended runs.	Task/schedule screenshot, successful execution timestamps, multiple recovery points.

9.1 Recommended validation commands

```
whoami
whoami /upn
ipconfig /all
nslookup aklab.local
nltest /dsgetdc:aklab.local
gpresult /r
Get-ADUser -Filter * -Properties Department, Enabled | Select-Object Name, SamAccountName, Department,
Enabled
Get-ADOrganizationalUnit -Filter * | Select-Object Name, DistinguishedName
```

10. Security, Limitations, and Improvements

10.1 Security principles applied

- Use least privilege: grant only the permissions required for a ticket or job function.
- Do not share administrator passwords; use individual accounts and appropriate delegated access.
- Test changes with the affected user or a dedicated test account.
- Avoid exposing passwords, private tokens, or unnecessary personal information in screenshots.
- Disable accounts during offboarding and review access/ownership before deciding whether deletion is appropriate.
- Keep scripts readable, validate input, handle errors, and avoid storing credentials in plain text.
- Restrict write access to the announcement JSON and PowerShell script to administrators; read access alone must not allow a domain user to change the displayed message or executable code.
- Treat a file recovery as a change-controlled operation: review the target version and avoid overwriting newer edits without confirmation.
- Validate Group Policy Preferences task creation separately from GPO application, and preserve relevant Event ID 4098 errors for troubleshooting.

10.2 Lab limitations

- The environment uses one domain controller and one client, so it does not represent a resilient enterprise deployment.
- A successful test on CLIENT01 does not demonstrate deployment across many workstations.
- The report is based on a learning lab; outcomes should be backed by dated screenshots and command output before being described as fully verified.
- Automatic employee ID assignment and broader software deployment are separate enhancements unless completed and validated.

- Production security, monitoring, backup recovery, patch management, and change control would require additional design and testing.
- Shadow Copies protect against some accidental edits but are not an independent, off-host backup and do not provide disaster recovery.
- The announcement state is per Windows profile; sign-in delivery is not a centralized audit of employee acknowledgement.
- The current demonstration uses one domain client and a small set of accounts; organization-wide delivery and scheduled snapshot reliability require additional evidence.

10.3 Future improvements

- Integrate a free-tier or trial ticketing platform if suitable, while keeping screenshots and documentation independent of the platform.
- Explore Microsoft Entra ID after the AD DS foundation is documented.
- Practise software deployment at scale through Group Policy or a management tool only after the single-client scenario is fully understood.
- Review employee ID generation for uniqueness, race conditions, and an authoritative source of truth before presenting it as production-ready.
- Implement a Multi-Factor Authentication system.
- Collect final unattended Shadow Copy run evidence, test a non-destructive file-version restore, and define retention requirements.
- Add an announcement archive, centrally reviewable delivery/acknowledgement records, and optional periodic checks for already signed-in users.

11. Skills Demonstrated and Lessons Learned

Skill area	Practical evidence / learning outcome
Active Directory administration	Creating and locating directory objects; understanding users, groups, OUs, and account status.
Windows domain troubleshooting	Investigating DNS, domain controller discovery, sign-in, and domain membership.
Group Policy	Linking policies, refreshing policy, and using gpresult to check effective application.
File and folder permissions	Investigating share and NTFS permission layers and validating access as the affected user.
PowerShell	Using AD cmdlets, pipelines, Select-Object, Format-Table, Export-Csv, error handling, and object counts.
Ticket documentation	Recording issue, investigation, root cause, resolution, validation, and evidence references.
Security awareness	Least privilege, careful offboarding, safe credential handling, and separation of local elevation from domain rights.
Problem-solving	Using error messages to identify line-continuation mistakes, incorrect group names, and missing validation.
File recovery and snapshots	Enabled and tested Shadow Copies for shared files; distinguished volume snapshots, Previous Versions, and independent backup.
Scheduled operations	Configured and diagnosed time-based snapshots and user logon-triggered tasks; checked triggers, results, and account contexts.
User communications automation	Created JSON-driven IT notices, PowerShell pop-ups, per-user seen-ID tracking, and GPO Preferences deployment; investigated error 0x80070005.

The main lesson from this project is that configuration alone is not enough. I need to verify the effective result from the client or affected account, record what changed, and preserve evidence that another person can review. Troubleshooting also requires checking the fundamentals—DNS, object names, group membership, permissions, and policy scope—before making additional changes.

12. Evidence Register

HD-001	PasswordResetRequest	Resolved	High	HD-001 PasswordResetRequest
HD-002	CannotAccessEmployeesSharedFolder	Resolved	High	HD-002 CannotAccessEmployees
HD-003	NewEmployeeEntryRequest	Resolved	Low	HD-003 NewEmployeeEntryRequ
HD-004	UnauthorizedSharedFolderAccess	Resolved	High	HD-004 UnauthorizedSharedFold
HD-005	UnableToAccessCompanyShareNetwork	Resolved	Critical	HD-005 UnableToAccessCompar
HD-006	ManagerAccess	Resolved	Medium	HD-006 ManagerAccess.pdf
HD-007	ManagerFolderAccessRequest	Resolved	Medium	HD-007 ManagerFolderAccessRe
HD-008	RequestingTemporaryLockdownOnAccount	Resolved	Low	HD-008 RequestingTemporaryLo
HD-009	ConfiguringNewTeamAndUsers	Resolved	Medium	HD-009 ConfiguringNewTeamAn
HD-010	IntroducingChangedAccountLockoutPolicy	Resolved	Medium	HD-010 IntroducingChangedAcc
HD-011	AssigningEmployeeID	Resolved	High	HD-011 AssigningEmployeeID.pd
HD-012	EnablingDisabledAccount	Resolved	Medium	HD-012 EnablingDisbaledAccoun
HD-013	UnidentifiedLoginAttempts	Escalated	Critical	HD-013 UnidentifiedLoginAttemp
HD-014	ConsecutiveFailedAuthenticationAttemptsB	Resolved	Medium	HD-014 ConsecutiveFailedAuther
HD-015	UnabletoaccesstheMarketingfolderunderCo	Resolved	High	HD-015 UnabletoaccesstheMark
HD-016	BackupOperatorUnabletoPerformFileBacku	Resolved	High	HD-016 BackupOperatorUnablet
HD-017	OffboardingEmployeeNoahWilson	Resolved	High	HD-017 OffboardingEmployeeNc
HD-018	DedicatedFolderBackupOperatorAccount	Resolved	Medium	HD-018 DedicatedFolderBackupC
HD-019	SoftwareInstallationForMarketingTeam	Resolved	Medium	HD-019 SoftwareInstallationForM
HD-020	AssignAUniqueEmployeeIDToJamieTaylor	Resolved	High	HD-020 AssignAUniqueEmployee
HD-021	DevelopPowerShellScriptsforActiveDirectory	Resolved	Medium	HD-021 DevelopPowerShellScript
HD-022	ConfigureFileVersionHistoryShadowCopies	Resolved	Medium	HD-022 ConfigureFileVersionHist
HD-023	AutomatedVersionHistory	Resolved	Medium	HD-023 AutomatedVersionHistor
HD-024	CentralisedAnnouncementSystem	Resolved	High	HD-024 CentralisedAnnounceme

Refer to this document if required. here

Appendix — Useful Validation Commands and Reports

Purpose	Command
Show current identity	whoami
Show user principal name	whoami /upn
Review IP/DNS configuration	ipconfig /all
Test domain DNS resolution	nslookup aklab.local
Locate a domain controller	nltest /dsgetdc:aklab.local
Review applied Group Policy	gpresult /r
List AD groups	Get-ADGroup -Filter * | Select-Object Name, SamAccountName
List OUs	Get-ADOrganizationalUnit -Filter * | Select-Object Name, DistinguishedName
List disabled accounts	Get-ADUser -Filter 'Enabled -eq $false' -Properties Department, WhenChanged
Count a collection	$MemberCount = @($Members).Count
Check Shadow Copies configuration	vssadmin list shadowstorage
List existing shadow copies	vssadmin list shadows
Review announcement task	Get-ScheduledTask -TaskName "AKLAB-IT-Announcement"
Inspect announcement task results	Get-ScheduledTaskInfo -TaskName "AKLAB-IT-Announcement"
Check per-user announcement IDs	Get-Content "$env:LOCALAPPDATA\AKLabIT\SeenAnnouncements.txt"

Purpose	File Name
Architecture Report	AKLAB_Architecture.txt
Server Report	AKLAB_Server_Report.html
Users Report	AKLAB_Users.html
Gpresult	GPResult.html

End of report